

Use of Transfer Learning for Affordable In-Context Fake Review Generation

Luis Ibañez-Lissen , Lorena González-Manzano , José M. de Fuentes , and Manuel Goyanes

Abstract—Fake content is a noteworthy threat which is managed by assorted means. This is a serious problem for online shopping platforms whose products can be affected by negative or positive reviews. Artificial intelligence is commonly applied for fake review generation, being transfer learning a promising approach to reduce training requirements. However, the feasibility of generating in-context fake reviews using transfer learning has not been explored yet. This paper analyses the suitability of a couple of transformers (T5 and BART) to generate realistic in-context fake reviews. Results show that 1) the diversity of generated reviews is comparable to existing works; 2) human-based detection is close to random; 3) just reviews generated with one of the used transformers can be detected with 38% precision; and 1 h of training and 8 k real reviews are needed to produce realistic fake reviews.

Index Terms—Transfer learning, in-context fake reviews, generation, transformers.

I. INTRODUCTION

NOWADAYS, the democratization of Big Data [1] and social media platforms have become a tool to analyze, share, watch, and read content. At the same time, the improvement of machine learning techniques and the fast content spread and

consumption are leading to a new public concern – the so-called fake contents.

Fake content refers to any type of media that has been modified or generated to fool users. Typically, they are used to spread misinformation [2] or to manipulate their perceptions [3]. The recent Ukraine invasion has shown one use of these contents in a socio-politic war environment – a “Deep fake” of president Zelensky claims Ukrainians to surrender [4], [5]. This highlights the importance of studying the different types of fake contents [6], [7], [8] as well as the methods to detect them [9], [10], [11], [12]. Among them, text-based fake contents are the hardest to detect, as their features depend on the context and language [13].

Fake reviews are a noteworthy text-based type of fake content. They have been effectively used to influence users [14], [15], [16]. Indeed, they have an estimated economic impact of \$152 billion.¹ Thus, they pose a major challenge to e-commerce platforms – they affect their operation but they are hard to detect [17].

In the early days, fake reviews were manually produced to attack specific targets [18]. However, modern Artificial Intelligence (AI)-based approaches have been applied in this regard. Two types of fake reviews can be identified – *out-of-context* and *in-context*. The former refers to generic fake reviews, whereas the latter is tailored for a particular target (e.g., a given restaurant). Yao et al. [18] first studied these concepts describing the attacker model. A recurrent neural network was trained to generate fake reviews which were lately manually modified to fit the context. Authors concluded that the results were good enough to fool automatic filters and even to pass as human-written reviews. To produce in-context fake reviews, Juuti et al. applied a neural machine translation approach [19] and Jin et al. introduced a transformer encoder-decoder [20].

Previous approaches rely on standalone AI techniques developed from scratch. However, transfer learning techniques allow leveraging previously trained systems that can be repurposed [21]. Since they may have been trained with assorted vocabularies and large datasets, their fake reviews could be more varied and natural-looking. From a cybersecurity point of view, these techniques may allow low-budget attackers to succeed, given the availability of tools such as Chat-GPT [22]. This kind of AI-generated reviews could be for instance, sold in the darkweb and emerging a new service of the underground

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¹<https://www.weforum.org/agenda/2021/08/fake-online-reviews-are-a-152-billion-problem-heres-how-to-silence-them/>, last access February 2024

economy [23], [24]. Previous approaches have addressed the generation of generic fake reviews. Salminen et al. [25] and Perez-Castro et al. [26] studied the use of transfer learning with GPT-2 for generating out-of-context reviews. The former produced a dataset of fake reviews for detection purposes, whereas the latter aimed to generate sentiment-preserving reviews. Nevertheless, the feasibility of generating in-context fake reviews has not been explored yet.

To address this issue, this paper focuses on the use of transfer learning techniques for generating realistic in-context fake reviews. As opposed to generic reviews, in-context ones enable targeted attacks. By characterizing this threat, novel detection techniques can be devised in future, as current efforts² are not enough. Our approach differs from current popular AI-based text generators such as Chat-GPT in a couple of ways. On the one hand, Chat-GPT is instructed not to produce fake reviews, so countermeasures are in place to avoid such a misuse. On the other hand, Chat-GPT has been trained to perform over generalist tasks while our approach bases on models trained to work over specific contexts. Therefore, the research questions and contributions of this paper are as follows:

RQ1. Can transfer learning be applied to in-context fake review generation?

- We leverage T5 and BART models using two well-known datasets – the Yelp restaurant reviews dataset and the one from Amazon related to electronic products.
- We measure their suitability against both automatic and human-based detection.

RQ2. Is it possible to produce high-quality reviews with affordable resources?

- We characterize the amount of time and real reviews needed to produce credible fake reviews.

Answering these questions is valuable for e-commerce platforms and customers as they help to understand the actual impact of fake reviews. In this way, this work encourages e-commerce platforms to enhance their review systems and create trustworthy services. In the same way, it raises customers' awareness to carefully read reviews. Indeed, the more knowledge in the field, the better the identification [27].

The remainder of this paper is as follows. In Section II the state-of-the-art on transformers, transfer learning and text generation is introduced. Section III describes the proposed approach and the addressed goals. The evaluation is presented in Section IV. Section V describes the related work. Finally, conclusions and future work are outlined in Section VI.

II. BACKGROUND

This section provides the main notions related to the proposal, namely transformers and transfer learning, text generation and their quality metrics.

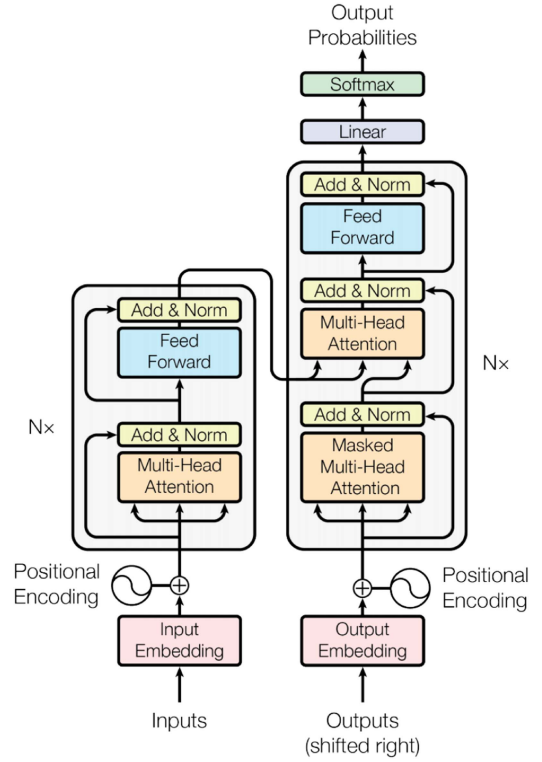


Fig. 1. Sequential model with Attention layer [32].

A. Transformers and Transfer Learning

Recurrent neural networks have been widely used for natural language processing [28], [29], [30]. They allow to input information in a sequential way, creating a set of internal states that can be used to solve a variety of tasks. However, their sequential nature makes them unsuitable for long texts. Therefore, attention mechanisms were introduced as a memory reservoir to improve their capabilities [31].

To improve the bottlenecks produced by the sequential input processing of RNNs, Vaswani et al. proposed *transformers*. Their suitability to capture semantic relationships in texts has been shown effective for other text-related tasks such as generation or classification [32]. Their architecture is shown in Fig. 1. They enhance the scalability by introducing a new positional encoding module and a variation on the attention mechanism. The left-hand side is the encoder, which takes a set of input embeddings and encodes them with the positional encoding module. This information is sent to the attention module and lastly to a feed-forward layer. Afterwards, it is shared with the decoder on the right-hand side.

The decoder is formed by a set of positional encoding and attention modules. They allow the model to extract features from the inputs. Then, such features serve as outputs for a final soft-max activation layer that determines the next most probable token.

The positional encoding Formula ((1) and (2)) introduced a novel approach which allowed to maintain the order of the elements of the input sequence while preserving the text meaning. The result is a matrix of embeddings which serves as an input

²<https://www.aboutamazon.com/news/polispy-news-views/amazons-latest-actions-against-fake-review-brokers#:text=Amazon%20invests%20significant%20resources%20including,from%20our%20stores%20in%20s2022..> last access February 2024

of the attention mechanism.

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}}) \quad (1)$$

$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}}) \quad (2)$$

Attention mechanisms were redesigned into a "Scaled Dot-Product Attention" which introduced the concepts of Query (Q), Vector (V), and Key (K). The original formula was reformulated as shown in (3), where Q, K, and V are embedding input matrices and are used to calculate the correlation between the input elements [32]. The higher the correlation, the higher the dot product of the matrices. Then, the model can extract intra-semantic features of the text at any distance.

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

1) *Architectures*: The initial proposal of transformers consisted on models divided into two parts – encoder and decoder. The encoder extracts semantic information from the inputs by applying a set of Self-attention layers. If a decoder exists, all this information is shared by using a cross-attention module. On the other hand, the decoder is in charge of getting the encoder information and its inputs which can be used for different purposes.

Examples of encoder-decoders are AlphaCode [33], Switch [34], T5 [35] or BART [36]. Later proposals decoupled the encoder and the decoder. For example, only the encoder was used to compress information and make classification tasks [37]. Similarly, unconditional text generation tasks were carried out with a decoder-only architecture [38].

2) *Transfer Learning*: Transfer learning in transformers is becoming a trend in the natural language processing field. It allows to take advantage of fine-tuning an already trained model for a specific task. In other words, a model pre-trained in a domain with a set of tasks is partially updated to minimize the loss in a new domain with other tasks [39] to obtain a new fine-tuned model.

One common method of fine-tuning that can be applied in transformers is the so called layer-wise learning rate [40]. It consists on setting different learning rates depending on the depth of the layer. Thus, higher learning rates are set on the outer layers and smaller in the deepest ones [37]. This is because outer layers tend to learn how to extract more general attributes and patterns while inner ones learn more task-specific characteristics.

B. Text Generation. Quality Metrics

There are several strategies to obtain generated text from the models. The main idea is to take the tokens of words in the vocabulary that better fit the context and desired criteria. Mathematically, it can be modeled as (4), where y is the distribution of tokens or words from the vocabulary x . The function arguments of the maxima ($\arg \max$) maximizes the most likely output.

$$y = \arg \max p(y|x) \quad (4)$$

The previous formula can be used alongside deterministic or stochastic strategies [41] which directly affects the quality of the generated texts. The most common deterministic algorithms

are Greedy search [42], which takes always the word with the highest probability, and Beam search [43] which keeps track of the different paths to choose the one with a better probability.

There are different sampling strategies among the stochastic algorithms. On the one hand, top-k selects a random word from a fixed-size distribution of k tokens. On the other hand, top-p selects a random token among those included in the probability distribution of value p .

There is no consensus in academia about which decoding strategy produces better results [44], [45] as there are many factors that may alter the quality. Beam search, with the proper parameters set-up, offers a more natural readable results, while top-k and top-p produce more diverse texts. They can be used jointly in a mixed strategy to reduce the sampling space.³ First, the probability distribution is adjusted, summing up to the top-p probabilities, and then selecting between the resulting top-k candidates, thereby avoiding the inclusion of low-ranked words.

1) *Metrics*: When producing texts, three metrics are at stake, namely Diversity, Flesch reading ease score and McAlpine Eflaw score. Each one is introduced below.

Diversity measures the lexical variety of the outputs produced by the models. It can be computed by calculating n-grams. Li et. al [46] proposed the *dist - n* metric, which calculates the number of distinct n-grams divided by the total number of n-grams (5). An n-gram is a contiguous sequence of n-items, e.g., words in this work.

$$Dist - n = \frac{Distinct\ Unique\ ngrams}{Total\ ngrams} \quad (5)$$

This metric does not measure the readability or how natural the text is. However, it is interesting as it may reveal how diverse are the generated in-context reviews. Complementarily, Uniq-n refers to the uniqueness of n-grams, meaning how many distinct n-grams are present in a given corpus.

The Flesch reading ease score measures the readability of the texts on a scale of 0 to 100. The higher the score, the easier and more natural the text is. It was created to help newspapers to write easier articles for native English speakers. It is described in (6) where TW is the total words, TS the total sentences and TSy the total number of syllables:

$$FleschScore = 206.834 - 1.015 * \left(\frac{TW}{TS}\right) - 84.6 * \left(\frac{TSy}{TW}\right) \quad (6)$$

Finally, McAlpine Eflaw is similar to Flesch but it measures the ease of a text for non-native English speakers. It scales from 1 to over 30, being 1 very easy to read. The desired value is between 15 and 25 [47]. The Formula (7) sums the number of words (W) and mini-words (M) and divides it by the number of sentences (S)

$$McAlpine\ Score = \frac{W + M}{S} \quad (7)$$

III. PROPOSED APPROACH

An overview of the proposal and the goals at stake are introduced in the following.

³<https://huggingface.co/blog/how-to-generate>, last access February 2024

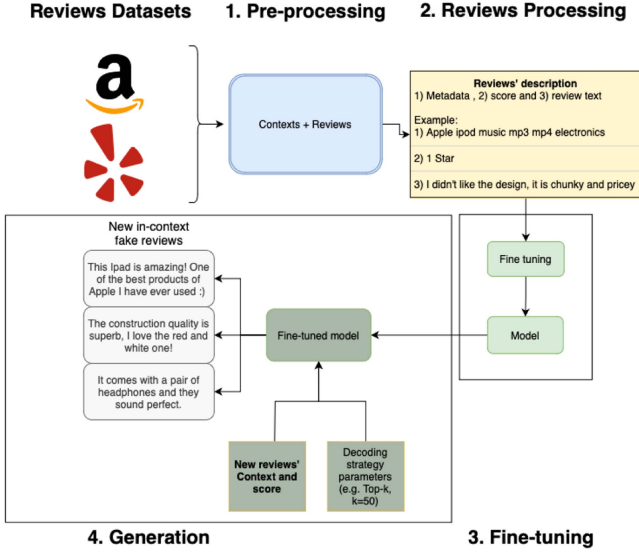


Fig. 2. Overview of the approach.

A. Overview

In this proposal, transfer learning techniques are used to fine-tune existing models. Recalling the architectures introduced in Section II-A, we opt for encoder-decoder ones. This is because encoder-only cannot produce outputs, whereas decoder-only cannot receive additional information (in our case, the context) to produce outputs. In particular, the transformers at stake are T5 [48] and BART [36]. They are state-of-art in encoder-decoder transformers and both have been designed with transfer learning tasks in mind. Indeed, they allow performing sequence-to-sequence tasks for text generation and they have been designed for executing general tasks. In particular, T5V1.1 version of T5 model is applied, which is a revision of the original version with some improvements such as the GEGLU activation function [35].

BART is suitable for in-context fake review generation. When producing text, it is able to process information from left-to-right and right-to-left contexts. Therefore, it allows to better keep track of the generated text according to the given context. On the other hand, T5 was the first transformer trained for multitasking in text. As in-context generation requires handling additional elements (i.e., metadata) beyond the generation requirements, such a multi-tasking flexibility is a relevant feature. Thus, our choices enable characterizing how a more generalist pre-trained model behaves versus a pre-trained model only in one task.

Our method is in-context for a couple of reasons. Firstly, the model is fine-tuned with reviews' datasets to directly focus on reviews. Secondly, additional metadata of the desired reviews are fed into the model in the generation phase, thus conditioning the model output.

A general overview of the approach is depicted in Fig. 2. During the pre-processing phase, the reviews from the datasets are extracted for further processing. After extraction, all text reviews are paired with their corresponding context and score and then batched for training. Once the training pipeline is ended, the fine-tuned model is saved for future inference tasks. After

training, the models generate new reviews given a new context and score. Each of these steps are detailed in the following.

1) *Pre-Processing*: The preprocessing is sub-divided in the following pair of phases, following [19], [20]:

- **Context processing phase:** A new string is created with the target score and the most characteristic fields available on the datasets to represent the context.
- **Review processing phase:** A “[start]” and “[end]” tokens are added to let the decoders know when to start or stop producing text. In addition, spaces between words and punctuation marks are included.

2) *Reviews Processing*: This step involves processing reviews to get three elements for each of them – the metadata, the score and the review itself, where metadata refers to the context information. In particular, both datasets contain different information that can be used as metadata to describe the target restaurant or product. For Yelp, the fields chosen to construct the context information are ‘name’, ‘address’, ‘city’, ‘state’, and ‘categories’. The latter is a general description about what type of food they offer. For Amazon electronics dataset, the fields ‘brand’ ‘title’ and ‘category’ are selected, where ‘title’ contains general information about the product.

Then, reviews are split by score from 1 to 5, creating a set of 5 data-frames (one per score). The idea is to later form a new global training data frame with the same number of reviews per score to avoid any kind of bias or data imbalance. On the final set for training, the same number of reviews of score 1, 2, 3, 4 and 5 are applied.

3) *Fine-Tuning*: The models are downloaded and fine-tuning is applied through the layer-wise learning rate technique (recall Section II-A2). All training data frames are passed to the models during training. Once the model has been trained during the specified number of epochs, the final version is accordingly labeled and saved for future inferences.

4) *Generation*: For generating reviews, new reviews’ metadata and score are introduced. In this way, the attacker chooses a context aligned with the provided metadata. Afterwards, parameters linked to the decoding strategy are specified and reviews are generated.

B. Goals

The main goals addressed in this paper are the following ones:

- **Realism:** The reviews must be legible, semantically correct, and related to the topic specified in the context. Humans must not be able to detect them as computer-generated.
- **Undetectability:** In-context fake reviews must be able to remain undetectable by automatic detectors.
- **Affordability:** The creation of reviews must be affordable in terms of training data and time.

IV. ASSESSMENT

The proposed approach is assessed through a series of experiments. Thus, the datasets and the experimental settings are described in Sections IV-A and IV-B, respectively. Afterwards, a preliminary data assessment is conducted (Section IV-C) to

TABLE I
LINGUISTIC FEATURES PER DATASET

	Electronics (Amazon)	Restaurants (Yelp)
Avg. sentence length	16.71	15.3
Lexical diversity	0.1	0.07
Avg. syllables per word	1.24	1.22
Polarity score	0.2	0.27

check whether the produced reviews are similar to the considered real reviews. The three goals at stake (recall Section III-B) are assessed in Sections IV-D, IV-E and IV-F, respectively. Lastly, a discussion involving the limitations of this work, the ethical concerns and open issues is presented in Section IV-G.

A. Datasets

Yelp⁴ restaurant reviews and Amazon reviews of electronic products datasets⁵ are considered. These datasets contain real information about users, reviews, businesses, images, etc. extracted from the corresponding websites.

Regarding the Yelp dataset, containing a total of 7 M reviews, a subset is created by selecting the category “Restaurant” and removing entries that contain ‘null’ values in any of the fields. After the pre-processing step (recall Section III-A1) it results in 472,468 reviews with their context.

This dataset was chosen in order to be in line with previous works [19], [20].

Concerning the Amazon Electronics Dataset, 7 M reviews with their context are obtained. The context description differs from the ones in Yelp, as they have different fields. While Yelp contains information about the address and the state the restaurants belong to, Amazon provides a short description of the products. This dataset was chosen to assess the quality of the models in a different scenario than the restaurant one, thus guaranteeing that results are not domain-specific.

A linguistic comparison of both datasets is presented in Table I. The NLTK library [49], a well-known tool to identify textual features, is used to compute the average sentence length, the lexical diversity, the average syllables per word and the polarity score. The latter indicates negative reviews for negative values and positive otherwise. Results are quite similar – just reviews related to electronics are a little bit longer. As a result, given the similarity between both datasets, their use in this proposal helps to ensure that there are not domain-specific features affecting the fake reviews generation process.

B. Experimental Settings

For training and inference with T5 and BART, the “Hugging-Face” API⁶ is used because it offers an easy way to manage and download models as well as flexibility when generating texts. In terms of computational resources, Google Colab pro⁷ is applied.

TABLE II
DECODING STRATEGIES PARAMETERS

	Top k	Top p	Beam Search	Mixed
Settings	k:50	p:0.7	size:10	k:50 p:0.8
Number of sequences	3	3	3	3
Temperature	-	-	-	1.2
Repetition penalty	-	-	-	1.2

It gives access to different GPUs (Nvidia K80, T4 and P100) and ensures a minimum of 16 GB of GPU RAM and 24 GB of RAM, and CPU is claimed to be an Intel Xeon.

There are different model sizes but for the sake of affordability, the smallest available ones are selected. In particular, the smallest T5 model has 60 M parameters while the BART base model is around 140 M parameters. Note that datasets and computing resources are publicly available, demonstrating that a malicious actor with low resources could replicate our approach to access state-of-art models to generate realistic in-context fake reviews. In this regard, our experimental tools will be released to interested researchers after a licensing agreement including ethical use restrictions.

Regarding the selection of hyper-parameters, the batch size is set by trial and error, testing until the training data is fitted into GPU memory. The optimizer is Adafactor [50] an evolution of the classical and well-known Adam optimizer [51]. It was used in the original proposal of the models [36], [52]. The learning rate is set to 0.01 for being a classical value that typically works with multi-layer architectures [53]. Following the same fashion, the multiplicative decay for the layer-wise learning rate technique is set to 0.95, which is considered to be a good starting point according to previous works [54], [55].

In terms of training, we perform partial checkpoints saves⁸ to save intermediate states of the models during 8 epochs to be in line with the rest of the works [19], [20].

For the text generation phase, we apply different techniques among those presented in Section II-B. These are set as the decoding strategy of the transformer at stake. In particular, we generate reviews with each of the models using beam search, top-k, top-p, and a mix of top-k and top-p. That is, a set of 941 reviews are generated per strategy and per model. Note that the number of generated reviews is chosen for comparison purposes against [19] and [20]. Table II shows a summary of used parameters, which are chosen for being the most common and best alternatives according to existing works [56].

The temperature allows to control the probability distribution – the higher, the less uniform and hence more diverse. The repetition penalty avoids the model using the same tokens all the time – the higher, the less repeated tokens and hence, more diverse. These parameters have been set after analyzing previous works [57]. It must be noted that high values for both parameters

⁴<https://www.yelp.com/dataset>, last access February 2024

⁵<https://jmcauley.ucsd.edu/data/amazon/>, last access February 2024

⁶<https://huggingface.co/>, last access February 2024

⁷<https://colab.research.google.com/signup>, last access February 2024

⁸https://pytorch.org/tutorials/beginner/saving_loading_models.html#saving-loading-a-general-checkpoint-for-inference-and-or-resuming-training, last access February 2024

TABLE III
QUALITY METRICS FOR THE PRODUCED FAKE REVIEWS

T5										
Yelp Restaurants						Amazon Electronics				
	Baseline	Top k	Top p	Beam search	Mixed	Baseline	Top k	Top p	Beam search	Mixed
Dist-1	0.09	0.14	0.10	0.07	0.11	0.20	0.16	0.12	0.09	0.13
Dist-2	0.48	0.58	0.38	0.17	0.53	0.69	0.61	0.43	0.25	0.57
Uniq-1	9,578	3,802	2,562	1,075	3,484	5,909	3,480	2,397	1,106	3,531
Uniq-2	105,283	14,729	9,319	2,672	15,657	27,700	12,545	8,218	2,765	14,727
Flesch	81.87	86.20	91.46	95.4	83.14	82.60	89.09	93.57	98.46	86.06
Alpine	19.14	17.43	14.43	14.01	22.29	15.61	16.51	14.12	11.08	21.43
BART										
Dist-1	0.09	0.061	0.054	0.059	0.062	0.20	0.069	0.062	0.07	0.062
Dist-2	0.48	0.176	0.139	0.144	0.19	0.69	0.239	0.20	0.199	0.22
Uniq-1	9,578	1,659.18	1,465.9	1,468.27	1810	5,909	2,038	1,769.46	1,205.2	2,253
Uniq-2	105,283	4,753	3,745.7	3,429	5,388	27,700	6,933	5,654	3,049	7,982
Flesch	81.87	89.78	88.23	87.03	89.59	82.60	91.11	91.27	94.02	90.91
Alpine	19.14	15.73	16.49	17.56	16.37	15.61	15.92	16.05	13	16.31

may reduce the coherence of the generated texts, so they must be set to keep a balance with the text diversity.

C. Preliminary Data Assessment

Before analyzing the goals of our approach, an initial analysis of the quality of the produced in-context fake reviews is carried out.

Table III shows the average values of the quality metrics (recall Section II-B1) of the produced fake reviews. The baseline refers to the results from the original datasets, which contain real reviews. For simplicity, the analysis is explained per model.

In what comes to T5, results vary depending on the decoding strategy but metrics tend to be close to the original. A mixed decoding strategy offers closer metrics to the original dataset. It can be seen that in terms of diversity (i.e., dist-1 and dist-2) our reviews are not only similar but even more diverse than Yelp. In the case of Amazon dataset, diversity (i.e., dist-1 and dist-2) is also close to the original dataset, specially with the top-k decoding strategy.

Readability metrics for both datasets are also close to the baseline, which may point out that the model is learning the attributes or the linguistic features of the datasets.

In the case of BART, the same decoding strategy produces less varied reviews than the ones from T5. However, they are to some extent similar to the original. As opposed to Dist-2, in Dist-1 the different decoding strategies are close to the baseline. However, readability scores in both datasets are even beyond the baseline.

Our results support the quality of the generated reviews and they are comparable with previous in-context works (see Table IV). Jin et al. [20] generated 941 reviews using both their method and the one presented in [19], calculating different diversity indicators by counting the number of unique uni-grams (uniq-1) and bi-grams (uniq-2). The average of the metrics from the reviews in the two contexts obtained with our T5 model outperforms the previous works in terms of diversity. BART model performs worse in the Dist-2 metric but improves a 20% on Dist-1. The same happens in Uniq-1 and Uniq-2 – T5 presents better results than any other work, while BART presents comparable results in terms of Uniq-1 but worse for Uniq-2. Thus, our approach leveraging T5 improves the previous efforts

TABLE IV
DIVERSITY COMPARISON AGAINST RELATED WORKS

	Juuti et al. [19]	Ji et al. [20]	Salminen et al. [25] (out of context)	Fine-tuned T5 (best / average)	Fine-tuned BART (best / average)
Dist-1	0.05	0.05	0.16	0.16 / 0.11	0.07 / 0.06
Dist-2	0.32	0.33	0.57	0.62 / 0.44	0.24 / 0.18
Uniq-1	1,578	2,088	4,512	4,292 / 2,679	3,226 / 1,708
Uniq-2	8,993	13,377	20,083	17,101 / 10,079	11,838 / 5,116

in terms of diversity, while BART provides quite comparable results.

Comparing with out-of-context proposals, Salminen et al. [25] results are close to ours. However, GPT-2 lacks the possibility of conditional inputs, such as our contexts.

D. Realism

To measure realism, a set of subjects were recruited through Amazon MTurk⁹ to determine their ability to spot fake reviews. A survey was carried out mixing real reviews and our fake reviews. 12 different surveys (6 per model) were prepared by picking fake reviews from the different training setups and mixing them with real reviews. 20 reviews (10 of each context) are included in two different exercises. In the first one, “Label as fake/real”, respondents are asked to classify whether each of the 5 reviews are fake or not. Thus, their random guess probability is 0.5 (out of 1), becoming a binary classification problem where each of the reviews are either fake or genuine. For the second exercise, called “Spot the fake”, respondents are asked to pick up the fake review between a list of 5. Thus, random guess is 0.2 in this exercise. It must be noted that respondents are given the same context information as the one provided to the considered models.

A total of 100 responses were recorded – 35 for BART and 65 for T5. 65% of the participants were from North America, 32% from Asia and 3% from Europe. 50 % have a Bachelor degree while 29% have a Master. Regarding the first exercise, results in Fig. 3 with respect to T5 show that humans’ accuracy to correctly label a review as real or fake is 0.51. BART shows almost a similar accuracy, that is 0.52. This means that humans cannot distinguish between real or fake reviews as this value is almost the same as the random guess.

Results in the second exercise confirm the same trend as in the first one (see Fig. 3). In this case, there is a mean of deviation of the 5% with respect to the random guessing baseline. Therefore, the produced reviews offer very little valuable information to spot them as fake.

Results taking into account each model and the amount of reviews used for training are also analysed. Table V shows

⁹<https://www.mturk.com/>, last access February 2024

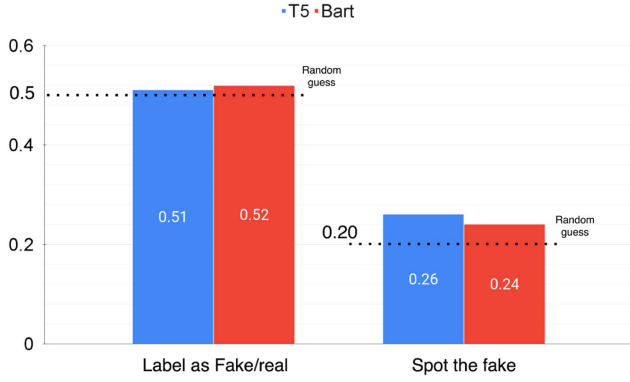


Fig. 3. Human-based detection accuracy.

TABLE V
ACCURACY OF HUMAN DETECTION PER MODEL AND TRAINING SIZE (LABEL REVIEWS / SPOT THE FAKE)

	10k	9k	8k	7k	6k
T5	0.53 / 0.38	0.53 / 0.25	0.53 / 0.27	0.56 / 0.08	0.47 / 0.25
BART	0.5 / 0.45	0.43 / 0.08	0.52 / 0.25	0.56 / 0.27	0.49 / 0.18

no clear correlation between the number of reviews used in the training phase and the accuracy on classifying whether the reviews are fake or genuine in none of the exercises.

E. Detectability

To the best of authors knowledge, there are few proposals specialized on detecting generated fake reviews. Most detectors are based on Deep learning with RNNs [58], [59], while others are starting to apply transfer learning classifiers [25], [60], GAN like set-ups [61] or multi-feature models [62] and show promising results, becoming state of art in the task.

To check if the generated fake reviews are good enough to fool detectors, the fine-tuned RoBERTa model is used [63]. RoBERTa has shown to be very powerful for text classification tasks [25]. Moreover, it is a fair choice as the architecture is different as the ones used to generate our in-context fake reviews. Firstly, the proper configuration of RoBERTa is ensured. To do so, the configuration and training reported in [25] is repeated with the same dataset (available online¹⁰), getting similar accuracy rates – 98.3% for training and 97.21% for testing. Once the model is properly reproduced, a total of 18,820, fake reviews were generated from each of our models (see Section IV-B), that is, 10 times 941 batches of reviews by the two datasets, Yelp and Amazon electronics. They were divided in batches of 941 in-context fake reviews and 941 real reviews, to analyze if the reviews were good enough to fool the detector.

As the quality of the fake review may be affected by the size of the training set, different sizes were chosen. In particular, two fractions of the dataset (namely 25% and 10%), and four fixed

TABLE VI
DETECTABILITY RESULTS

	T5									
	Yelp Restaurants					Amazon Electronics				
	25%	10%	10k	8k	6k	25%	10%	10k	8k	6k
Precision										
Beam search	0.39	0.45	0.48	0.60	0.56	0.36	0.18	0.17	0.19	0.55
Top k	0.51	0.50	0.50	0.55	0.51	0.19	0.22	0.26	0.24	0.40
Top p	0.87	0.43	0.85	0.90	0.90	0.5	0.43	0.48	0.47	0.64
Mixed	0.36	0.49	0.45	0.45	0.26	0.36	0.13	0.26	0.21	0.45
Accuracy										
Beam search	0.32	0.50	0.77	0.73	0.76	0.53	0.49	0.55	0.56	0.75
Top k	0.49	0.48	0.73	0.70	0.73	0.51	0.62	0.59	0.57	0.66
Top p	0.58	0.58	0.96	0.91	0.92	0.46	0.46	0.7	0.66	0.77
Mixed	0.48	0.48	0.69	0.68	0.64	0.59	0.60	0.59	0.55	0.68
	BART									
	Yelp Restaurants					Amazon Electronics				
	25%	10%	10k	8k	6k	25%	10%	10k	8k	6k
Precision										
Beam search	0.99	1.0	0.92	0.97	0.97	0.82	0.78	0.41	0.43	0.44
Top k	0.99	0.99	0.96	0.97	0.98	0.87	0.83	0.87	0.84	0.86
Top p	0.98	1.0	0.91	0.97	0.98	0.89	0.88	0.88	0.85	0.89
Mixed	1.0	1.0	1.0	1.0	1.0	0.89	0.88	0.87	0.86	0.90
Accuracy										
Beam search	0.68	0.72	0.92	0.95	0.93	0.57	0.46	0.68	0.69	0.69
Top k	0.68	0.7	0.94	0.97	0.93	0.57	0.59	0.91	0.9	0.91
Top p	0.68	0.7	0.91	0.95	0.94	0.61	0.55	0.91	0.9	0.92
Mixed	0.70	0.72	0.97	0.96	0.96	0.57	0.43	0.91	0.9	0.92

sizes (from 10 k to 6 k) were chosen. They were experimentally set to further address the affordability goal (Section IV-F).

For this analysis, our 941 fake reviews were mixed with a set of the same size of random genuine reviews. This was repeated 10 times. Table VI summarizes the average results (between 0 and 1), in which values equal or lower 0.5 are highlighted – detection is not considered as effective enough. Precision reflects the capability to identify fake reviews accurately. In the case of T5, when using beam search, top-k, and a mixed strategy during decoding, RoBERTa is not able to detect fake reviews in most settings neither on Yelp nor in Amazon. Indeed, it reaches an average precision of 0.38. By contrast, the top-p decoding strategy is easier to detect. As accuracy points out, in-context fake reviews affect the overall performance of the detector, being especially decreased using the mixed strategy.

On the other hand, in the BART model, under the same decoding setup, the detector spots the generated fake reviews. Neither decoding strategy produces convincing texts for the detector, correctly labeling most of the reviews. Moreover, it does not impact the overall accuracy of the RoBERTa model.

Comparing humans and automatic detectors, Table V shows that there is no relation between the training size and the accuracy of humans to detect fake reviews. By contrast, detectability of the models depends on the amount of training size and the decoding strategy. This points that there are patterns easily understandable by automatic detectors but not for humans. Indeed, based on EU recommendations¹¹ for fake content detection by humans, the detection process bases on verifying content, medium or

¹⁰<https://osf.io/tyue9/>, last access February 2024

¹¹[https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA\(2017\)_599386](https://www.europarl.europa.eu/thinktank/en/document/EPRS_ATA(2017)_599386), last access February 2024

TABLE VII
EXAMPLES OF REVIEWS FOR DIFFERENT TRAINING DATA

Context	Panda King Gretna LA Restaurants Chinese	
Score	Score 1	Score 5
7k	This place is a little pricey for what you get and it's not cheap but worth the money! The food was really good too! I will definitely order from here again.	I've been to this Panda express for years and have never had a bad experience here! The food was great but the service is horrible. We got up in 20 minutes before we left, so it took forever
8k	I was a little disappointed with the food here and it's not worth it! We had to wait over an hour for our entree and were told that we would only order one dish because of the service.	I was in town for the first time today and had a good experience at this location! The food is delicious but the service can be very slow when we come here. They have some pretty simple dishes which are usually quite tasty.
25% of dataset	I went during lunchtime, ordered the spicy shrimp and it was served with rice which tasted like they put too much salt in their food. It had some good flavor but not that great after all :(	The food was amazing for what it charge, the service is great and very friendly I really recommend this restaurant! You also will enjoy it here as they have an excellent selection of chinese foods including hot tea's to pair with you sweet or hot deliciousness from Panda king

authors, which can be correct in a fake review. By contrast, the effectiveness of automatic detectors is influenced by internal patterns learned in a training process.

F. Training Affordability

The affordability analysis focuses on studying the amount of training data to produce realistic reviews and the required time. Each issue is addressed separately.

1) *Training Data*: As shown in the detectability analysis, no strong correlations appear between the training size and the accuracy of detectors. However, it was still unclear whether the positive and negative semantics were in line with the purported scores.

A manual process was initially carried out to analyse the impact of the number of reviews on the quality of the generated text. Table VII shows some examples of reviews generated by T5 over the Yelp dataset with a given context, varying the number of reviews used for training. Positive comments are colored in green, negatives in red and words that match the context are highlighted. It is noticed that applying a training set smaller than 7 k reviews, the model is not able to correctly represent the target score. The model tries to insert some words of the given context in the review but it fails. Moreover, texts are not consistent and show some contradictions.

However, to quantify the amount of required training data, RoBERTa model was used to predict if the score of a given review is either 1 or 5. To fine-tune the model, 500 k reviews with a score of 1 and 500 k reviews with a score of 5 are collected from the Yelp dataset. The 1 M real reviews were divided into training (90%) and testing (10%), achieving a training F1-score of 99%.

After training, 941 fake reviews were generated per score (1 and 5) and training quantity (6k, 7k, 8k, 9k, 10k, 10% and 25% of dataset, recall Section IV-E). RoBERTa was applied to check the consistency between the sentiments and words, and the given score. Fig. 4 shows the success rate for both datasets and models. Using 6 k reviews, T5 is not able to generate good fake reviews for restaurants, but it is much better for electronic products. One reason could be the semantics and linguistic features of the Amazon dataset are simpler and easier to understand to the

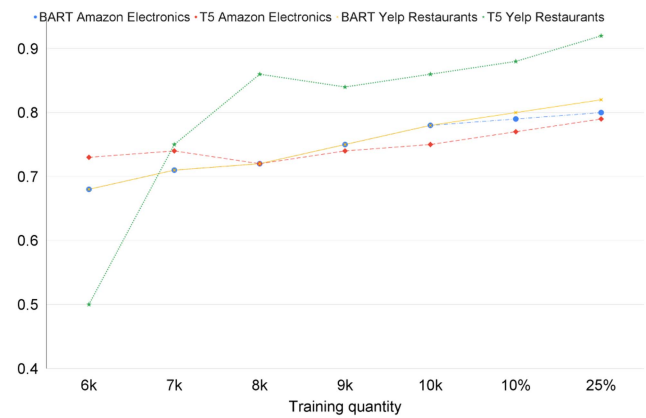


Fig. 4. Consistency between sentiments and score for different training sizes.

TABLE VIII
TRAINING TIMES (IN HOURS)

	25%	10%	10k	9k	8k	7k	6k
T5 Restaurants	131.86	37.6	1.27	1.15	1.05	0.91	0.85
T5 Electronics	44.8	19.3	1.13	0.98	0.9	0.73	0.59
BART Restaurants	15.46	5.9	1.25	1.12	1	0.86	0.73
BART Electronics	42.8	16.46	0.8	0.71	0.62	0.54	0.468

model. In the case of BART, results are quite similar for all training sizes, with 0.74 of success when the number of training data is 7 k. Remarkably, a sensible boundary is 8 k, as the percentage of correctly classified instances (that is, texts and scores consistent with sentiments) is 0.86 for T5 in the domain of restaurants and 0.75 in the remaining cases.

2) *Training Time*: The time required for training the models is shown on Table VIII. Considering that 8 k reviews has been shown to produce realistic fake reviews which are consistent to their score, the analysis is focused on that particular setting.

We observe some linearity in the training time of our approach, correlated to the amount of reviews used for training. In the case of 8 k training, both T5 and BART take less than 8 minutes per epoch (i.e., around one hour in total) to produce suitable fake reviews.

While the work in [20] does not provide the training time needed for their approach, [19] claims to need around 9.6 hours. In most cases our approach improves this number and it has been shown that our reviews are better than theirs in terms of realism.

G. Discussion

There are three main remarks to complete the analysis presented herein – limitations, open issues and ethical concerns. Each one is presented separately in the following.

1) *Limitations*: Four main limitations can be found in our work. First, all training datasets and pre-trained models are in English and have been trained over English corpus. Consequently, our models can only produce texts in this language, limiting the scope. There is an important lack of research on NLP regarding other languages. This is important if this technology wants to be widespread.

Second, this work applies T5 and BART but there are more encoder-decoder pre-trained transformers suitable for this task. In addition, the smaller version of the previous models has been used but bigger instances may lead to better results.

Third, used datasets, Yelp, and Amazon electronics, may contain a percentage of fake reviews, introducing bias. Despite Yelp uses some filtering algorithms to discard suspicious reviews (around the 16%) [64], some of them may pass the filters. However, we assume this quantity is relatively small compared to the whole dataset and then, it does not interfere on the overall performance of the detector. The same reasoning applies to Amazon, who claims to "stop millions of suspicious reviews before they're ever seen by customers" [65].

Lastly, in terms of the impact on fake review detectors, our results are limited. It has been shown that it is possible to produce texts that are good enough to fool one detector. However, commercial detectors may outperform the one selected herein. Anyway, this must be regarded as an arms race – GAN-like approaches [61], [66], used jointly with our in-context fake review generation technique could improve the results.

2) *Open Issues*: There are a plethora of open issues and challenges related to fake reviews detection. In the following, challenges from different points of view are outlined:

- E-commerce: finding effective and immediate review processes for the comments posted in the platforms.
- Academia: finding effective automated countermeasures and detection systems that break the equilibrium between generators and detectors. As of today, the Academia is trapped in a mouse-catching mechanic – once better defenses are found, new attacks are proposed. Thus, there is a challenge in finding detectors that do not rely on learning from static datasets but on understanding the evolution of fake reviews over time.
- Providers: finding new ways to detect bad practices, such as abusing systems to generate fake reviews.

A common issue in all the views is the actual nature of the reviews. Online reviews are on its essence short, subject to humor and feelings which are often difficult to understand. For instance, the fuzzy line between humor and irony is quite thin and the short nature of the texts does not help. In this vein,

researchers are already trying to solve the problem of language disambiguation [67].

3) *Ethical Concerns*: In addressing the ethical concerns at hand, the purpose of this paper is to point out the rising problem of fake reviews given the possibility of generating affordable realistic reviews. The vertiginous pace of the development of large language models are worsening the problem. Therefore, this paper calls for new developments to detect this threat against online shopping. In this vein, this research has been authorized by the Research Integrity Board of Universidad Carlos III de Madrid.

V. RELATED WORK

In the area of targeted in-context fake reviews generation, inspired by [68], Lipton et al. [69] propose a RNN specialized in generating conditional texts given descriptions or sentiments trained on the BeerAdvocate [70] dataset. They compare a set of architectures to find one that they maintain the contextual information while generating sequences of characters. They found that state of the art in encoder-decoder architectures was not good enough to perform language generation at the character level because the contextual information got lost when generating long text sequences. To overcome these issues, they design a character-level LSTM network that concatenates contextual information along the hidden states produced during the generation phase. This approach generates convincing results but keeping the contextual information may result in redundancy and a waste of resources.

Yao et al. [18] continue this research direction. They train a RNN under the premise that they are good enough to generate template-like reviews that could be further processed to fit the specific products. The model is trained over the Yelp dataset [71] and then, the text is analysed with Wordnet [72] to find relevant words in the templates. Once detected, words are swapped by others words more relevant to the specific context, reaching good results.

In [19] authors test the suitability of taking a neural machine translation approach on the same Yelp dataset. They use a LSTM RNN encoder-decoder with an attention module, where they encode the contextual information with the encoder and produce text with the decoder. They realize the lack of variability and originality of the targeted reviews when using Greedy decoding strategies. Thus, they introduce what they call "Bernoulli penalties", finding that adding little penalties to random subsets of words contributes to generating texts with natural flow.

Still in the review generation context, the use of a transformer model alongside a GRU encoder is proposed in [20]. They follow the same training approach than the previous work [19] by pairing the contexts with the corresponding reviews. Worried about the lack of metrics for measuring diversity and novelty during text generation, they define a new metric, referred to as DMet.

Regarding transfer learning, [73] presents a survey on the topic, where the different tasks that can be performed are classified based on these techniques. It is especially interesting for our research since given a set of discrete attributes (Words,

TABLE IX
RELATED WORK ANALYSIS

	Lipton <i>et al.</i> [70]	Yao <i>et al.</i> [18]	Juuti <i>et al.</i> [19]	Jin <i>et al.</i> [20]	Salminen <i>et al.</i> [25]	Chiang <i>et al.</i> [62]	Perez-Castro <i>et al.</i> [26]	Shawon <i>et al.</i> [67]	Knoedler <i>et al.</i> [76]	Our work
Dataset	Beer advocate	Yelp	Yelp	Yelp	Amazon	Yelp and Amazon	Yelp, Amazon, Turk, TripAdvisor	Own dataset	RealSelf	Yelp and Amazon
Training samples	250k	617K	2.9M	3.5M	40K Amazon top 10 categories	Non specified.	34,700 (in total)	6,014 fake/authentic reviews	150 real, 150 fake reviews	472k (Yelp) and 5.7M (Amazon)
Technique	RNN	RNN LSTM	RNN + Attention	Transformer + GRU	Transfer learning with GPT-2	Reinforcement learning	Transfer learning with GPT-2	GAN	ChatGPT prompt	Transfer learning
Year	2015	2017	2018	2020	2022	2023	2023	2023	2024	2024
Affordability	×	✓	×	×	×	×	×	✓	×	✓
Realism	✓	×	✓	✓	×	×	✓	×	✓	✓
Diversity	×	×	✓	✓	×	×	×	×	×	✓
Undetectability	×	×	×	×	✓	✓	✓	✓	✓	✓
In-context	✓	✓	✓	✓	×	×	×	×	×	✓

categories, metadata) the goal is to control the generated models. One of the most relevant models working on this area is CTRL, designed by Keskar et al. [74] who demonstrate the ability to control the output of the models conditionally through control codes.

Inside generation with transfer learning, Salminen et al. [25] propose to fine-tune the GPT2 decoder-only transformer and the ULMFiT model [40] to generate out-of-context fake reviews. Perez-Castro et al. [26] fine-tuned GPT2 to generate sentiment-preserving reviews. Both approaches tackle the problems from an out-of-context point of view so there is no way to specify what has to be generated. Quite recently, S. Knoedler et al. [75] study the use of ChatGPT to generate patient reviews in plastic and aesthetic surgery. They use the ChatGPT prompt to generate reviews and statistical tests analyse differences among real and fake reviews, together with a survey to 30 participants. By contrast, our work takes advantage of conditional text generation in order to produce tailored reviews to attack specific products or restaurants with a given language, score or sentiment.

Other recent strategy to generate fake reviews is the one followed by Chiang et al. [61]. They use a swapping word strategy to attack recommender systems. However, their methodology is based on text alterations or modifications via reinforcement learning instead of pure text generation as ours.

In addressing the problem of improving fake reviews generators, Shawon et al. [66] successfully use a decoder-only transformer in a GAN-like setup. They aim to fine-tune fake review generators to create better detectors and producers at the same time. Our work confirms that current models are good enough to fool them, in line with [26].

A summary of this analysis is depicted in Table IX. In line with existing works and for comparison purposes we use Yelp dataset, as well as Amazon Electronic dataset. The latter also serves to avoid domain-specific results. Moreover, the size of datasets is bigger than most approaches. Concerning the applied technique, similar to some of the latest works, we apply transfer learning. However, to the best of our knowledge we are the first ones to create in-context fake reviews while considering affordability, realism, diversity and undetectability.

VI. CONCLUSIONS AND FUTURE WORK

In this work, the suitability of transfer learning approaches for creating in-context fake reviews is addressed. Two state-of-the-art models have been applied over different domains, measuring diversity, realism, detectability and affordability. They have been assessed using real-world data and detectors, including human subjects. Results have shown that this technique may lead to producing realistic fake reviews that cannot be easily spot neither by humans nor automatic detectors.

All in all, our proposal is competitive in comparison with the state-of-the-art. Remarkably, the proposed approach requires fewer data and takes less time to generate convincing reviews. The characterization of this threat is useful to foster further research in the detection of fake reviews.

Regarding future work, it is interesting to study the application of these techniques in other encoder-decoder transformers. Additionally, new approaches to detect fake reviews must be devised paying special attention to the need of evolving with time. For example, the use of lifelong learning techniques could be a promising area of research in this regard. One possibility, indeed, could be to adapt the concept of honeywords [76] as a watermark. In this case, a pool of fake reviews with that mark would be created. When a bad actor tries to submit a similar review, it would allow taking protective actions. Moreover, the use of datasets in other languages and new contexts are open research lines. It also includes the use of self-rewarding pipelines using fake reviews generators in line with works proposed by [77] to explore whether generators can use other LLMs to improve themselves.

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